

of the advocacy. Attitude changes based on extensive elaboration of issue-relevant arguments (central route) are postulated to be more persistent, resistant, and predictive of behavior than attitude changes based on simple cues (peripheral route). In the next chapter we discuss methodological factors relevant to testing the ELM and in the remainder of this volume we present the empirical evidence for the ELM.

Chapter 2

Methodological Factors in the ELM

Introduction

In the preceding chapter we outlined the postulates of the Elaboration Likelihood Model. In the remainder of this volume we will review the evidence for the ELM. Before turning to this, however, it would be useful to cover some general methodological issues that arise in testing the ELM. We will first briefly address the procedures we have employed to assess attitudes and attitude change in our experiments. Then we will discuss procedures for assessing and operationalizing the three basic constructs in the ELM: argument quality, peripheral cues, and message elaboration. Finally, we discuss individual differences in the likelihood of message elaboration.

Assessing Attitudes and Persuasion

As noted in Chapter 1, we use the term "attitude" to refer to a general favorable, unfavorable, or neutral evaluation of a person, object, or issue. "Persuasion" refers to any effort to modify an individual's evaluations of people, objects, or issues by the presentation of a message. Importantly, our definitions identify the attitude concept with a general evaluative dimension rather than with specific cognitive, affective, or behavioral responses. This is not intended to diminish the importance of cognition, affect, or behavior in attitudes and persuasion, but rather these constructs are accorded independent conceptual status, so that their functions as antecedents and consequences of attitude change may be examined separately.

Ever since Thurstone (1928) introduced the first attitude scale, attitude measurement techniques have generally focused on assessing a global evaluative reaction to some stimulus. The classic scaling techniques of Thurstone (1928) and Likert (1932), techniques for measuring emotional or behavioral responses (cf., Lemon, 1973), and the more popular and

straightforward semantic differential (Osgood, Suci, & Tannenbaum, 1957) and single-item rating scales (cf., Cacioppo, Harkins, & Petty, 1981) are all designed to gauge how much one likes or dislikes, or appraises favorably or unfavorably, a stimulus. In research described in the subsequent chapters, we have generally employed multi-item semantic differentials with bipolar adjectives from the evaluative dimension serving as anchors (e.g., good/bad) and/or single-item rating scales (e.g., 1 = "agree completely," 9 = "disagree completely") on which subjects respond to evaluative statements. When both types of scales have been used, the responses to each are standardized and aggregated to obtain an overall index of people's attitude toward the target stimulus (see Figure 2-1 for example).

To mitigate the effects of response sets (e.g., acquiescence, social desirability), the ends of the scales denoting positive and negative attitudes are either counterbalanced or determined randomly. Also, a cover story is employed so that subjects are unaware that the study concerns their persuasibility. In some experiments, for example, subjects are told that the psychology department is cooperating with the School of Journalism in evaluating radio editorials that were submitted by colleges and universities throughout the country (e.g., Petty, Cacioppo, & Heesacker, 1981). The subjects are instructed that their task is to assist in rating the broadcast quality of selected editorials. An important attribute of the cover story is that it not draw undue attention to the subjects' attitudes toward the persuasive

1. Rate how you feel about requiring college seniors to take a comprehensive exam in their major as a requirement for graduation on the scales below.

COMPREHENSIVE EXAMS ARE:

bad	-4	-3	-2	-1	0	+1	+2	+3	+4	good
unfavorable	-4	-3	-2	-1	0	+1	+2	+3	+4	favorable
wise	+4	+3	+2	+1	0	-1	-2	-3	-4	foolish
beneficial	+4	+3	+2	+1	0	-1	-2	-3	-4	harmful

2. To what extent do you agree with the proposal requiring college seniors to take a comprehensive exam in their major before graduating?

do not agree	1	2	3	4	5	6	7	8	9	10	11	agree
at all												completely

Figure 2-1. Example attitude scales. These were the questions used to measure attitudes toward senior comprehensive exams in the Petty & Cacioppo (1979b, Experiment 2) study depicted in Figure 4-2. Similar attitude assessments were employed for the other studies reported in this volume.

appeal as a focus of study. After hearing a specially prepared persuasive communication, subjects are told that because their personal views on the desirability of the recommendation might influence the way they rate the broadcast quality of the tapes, a measure of their own opinion on the issue is desired. Subjects then indicate their postcommunication attitude using the scales (e.g., semantic differentials) provided. This procedure supplies individuals with a justification for reporting their attitudes toward the recommendation while attenuating the biasing influences of suspicion and social desirability.

Initial Attitude Changes

In most experimental studies of persuasion, the theoretical question of interest concerns the *relative* rather than the absolute effects of various factors on attitude change. Consider, for example, an experiment designed to examine the effects of Factor-X (e.g., message repetition) on persuasion. All subjects might be exposed to the same message in the same persuasion context, but some would be assigned at random to a condition in which there is a low level of Factor-X, others to a condition in which there is a moderate level of X, and others to a condition in which there is a high level of X. Postcommunication measures of attitudes provide information about the relative effects of Factor-X, since successful random assignment to conditions assures that on average, subjects in each condition begin with the same attitudes toward the recommendation. These experimental conditions do not provide information about what levels of Factor-X led to *absolute* attitude change, however, because it is possible that one or more of the levels of X failed to elicit significant attitude change.

For instance, if the results depicted in the top panel of Figure 2-2 represent the postcommunication attitudes obtained in this hypothetical study, then one could infer that a moderate level of Factor-X elicits more support for the recommendation than low or high levels of Factor-X. However, this relative result could have occurred because: (a) low and high levels of X produced *no* change, but moderate levels produced positive change, (b) all levels of X produced positive change, but moderate levels produced the most change, (c) moderate levels of X produced no change, but low and high levels led to change *away from* the position advocated (boomerang), or (d) all levels of X produced negative change, but moderate levels produced the least negative change. When these "absolute" questions are of theoretical interest (or when subjects cannot be assigned randomly to all conditions, as when studying recipient factors), we have utilized either pretests or external-control groups. A pretest allows one to determine whether significant changes in attitudes occur between it and posttest, but pretests can have undesirable effects such as committing subjects to their initial attitudes or sensitizing subjects to the focus of the study. Separating the pretest from the experimental study and varying the appearance of the attitude measures can minimize these

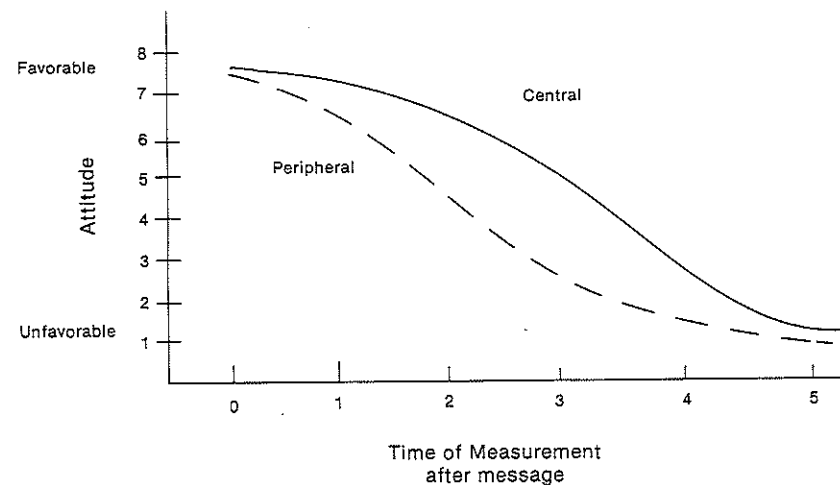
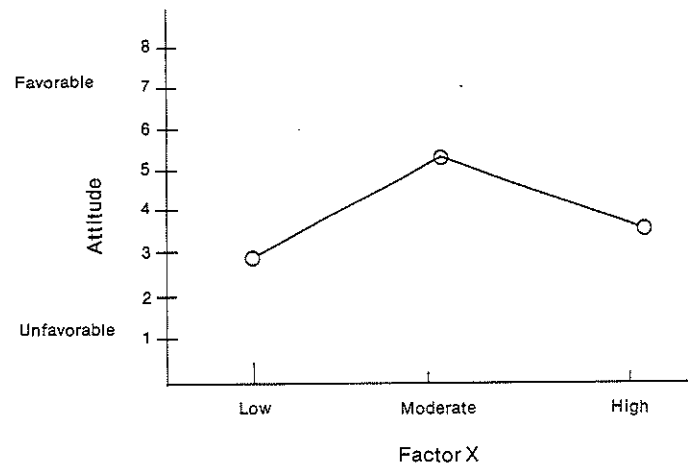


Figure 2-2. *Top panel*—effects of a hypothetical factor on attitudes (see text for discussion). *Bottom panel*—hypothetical decay functions for attitude changes induced via the central and peripheral routes (see text for discussion).

undesirable effects (Carlsmith, Ellsworth, & Aronson, 1976; Insko, 1967; Petty & Brock, 1981).

An alternative to using pretests that is less plagued by sensitization and carryover effects is to use external-control groups. For example, imagine that some subjects in our hypothetical study of Factor-X heard the experimental instructions but indicated their attitude toward the issue prior to their exposure to X. Comparisons of the attitudes expressed by these individuals with those expressed by individuals in each of the experimental conditions (e.g., using the Dunnett procedure) furnishes information regarding the absolute effects of each level of Factor-X on attitude change.

Attitude Change Over Time

Simply determining whether or not attitude change occurs does not in itself reveal whether the central or the peripheral route was traveled, since both routes can yield similar changes immediately following an individual's exposure to a persuasive communication. Recall that under the central route, recipients' susceptibility or resistance to persuasion is determined by their scrutiny and evaluation of the merits of the arguments for a recommendation. These "arguments" can be provided externally or generated internally. Thus, as noted in Chapter 1, under the central route, recipients' thinking about issue-relevant information is viewed as being the most direct determinant of the direction and amount of attitude change produced, and the attitude change induced via this route is posited to be relatively permanent, resistant to counterpersuasion, and predictive of behavior. Under the peripheral route, however, attitude change is viewed as resulting from pairing the recommendation with a simple cue that either alters recipients' moods directly, or provides them with the basis for a simple inference as to the appropriate attitude. Changes induced via this route are postulated to be relatively temporary, susceptible to counterpersuasion, and less predictive of behavior because, *ceteris paribus*, the attitude is associated with a cue that is more easily dissociated from the recommendation than are personally relevant beliefs and thoughts that are central to the merits of the position advocated.

It would appear then, that one way to assess whether changes are central or peripheral is to examine the consequences of the attitude changes. For example, one of the fundamental differences in the consequences of the two routes to persuasion is the temporal persistence of the induced attitude changes. It should be noted, however, that assessing this consequence may be rather difficult. Since the exact decay functions of attitude changes induced by the central versus the peripheral routes are not known, administration of one immediate and one delayed posttest does not provide a strong test. The bottom panel of Figure 2-2, which depicts hypothetical decay functions for central and peripheral attitude changes, illustrates the

problem. No differences in the delayed attitude measure would be expected if the second assessment were made too early, in which case peripheral persuasion might not yet have decayed. No differences would likewise be expected if the second assessment were made too late, in which case central persuasion might have already decayed. Moreover, because researchers lose control over extraneous factors when subjects leave the laboratory, the error variance associated with follow-up attitude assessments can be large, rendering the tests insensitive. Furthermore, repeatedly assessing the same individuals' attitudes toward a recommendation can itself exert an extraneous effect, such as committing individuals to their attitudinal positions or causing them to rationalize their reported attitudes. Finally, if different groups of individuals are tested at varying points in time following the persuasive communication, the number of subjects required to conduct studies can become prohibitively large.

Many of these cautionary comments (and even additional ones) are, of course, applicable to assessing the other consequences of the route to persuasion. For example, in assessing the resistance to counterpersuasion of attitudes newly formed via the central versus peripheral routes, what kind of countermassage is optimal? If a countermassage contains both strong arguments and strong peripheral cues, both groups may retreat from their new positions, but for different reasons. Likewise, in assessing attitude-behavior consistency, what behavioral criterion is appropriate? If the behavior tested is too easy, all groups may comply. If too difficult, very few will go along. Testing multiple behaviors may be necessary (cf., Ajzen & Fishbein 1977). Because of the potential difficulties of examining in a simple and efficient manner the postulated consequences of attitude changes induced via central versus peripheral processes, an alternative assessment procedure for distinguishing these two routes is clearly desirable.

One key to the solution of this problem lies in examining the distinctive underlying *processes* postulated by these two routes. Persuasion resulting from the central, in contrast to the peripheral route, is posited to be accompanied by a more careful and cognitively demanding examination of the merits of the recommendation. This difference should be discernible during the same session that subjects are exposed to the persuasive communication. In the following sections we examine several tactics used to assess the route to persuasion.

Assessing the Basic Constructs in the ELM

As we detailed in the last chapter, the ELM holds that variables can affect the amount and direction of attitude change in one of three ways: (a) serving as persuasive arguments, (b) serving as peripheral cues, or (c) affecting the extent or direction of argument elaboration (see Postulate 3). In this section,

we define in greater detail these three constructs in the ELM and we discuss operationalizations of each. As noted above, if we can distinguish attitudes that are based on argument elaboration rather than peripheral cues, we will distinguish the central and peripheral routes to persuasion. In subsequent chapters, we discuss how many of the typical source, message, recipient, channel, and context variables manipulated in the accumulated persuasion research can be understood in terms of this three-part categorization and the central and peripheral routes. In Chapter 7, we return to the postulated consequences of the two routes to persuasion.

Argument Quality

One of the least researched and least understood questions in the psychology of persuasion is: What makes an argument persuasive? As we noted in Chapter 1, literally thousands of studies and scores of theories have addressed the question of how some extra-message factor (e.g., source credibility, repetition, etc.) affects the acceptance of a particular argument, but little is known about what makes a particular argument (or message) persuasive in isolation. In fact, the typical persuasion experiment employs only one message and examines how some extra-message factor affects acceptance of the message conclusion. Furthermore, studies that do include more than one message often do so for purposes of generalizability across topics, not because the messages are proposed to differ in some theoretically meaningful way (e.g., Hovland & Weiss, 1951). There are, of course, notable exceptions to our generalization. For example, a few studies have manipulated the comprehensibility or complexity of a message (e.g., Eagly, 1974; Eagly & Warren, 1976; Regan & Cheng, 1973), mostly to test McGuire's (1968) information processing model, but even these studies were not aimed at uncovering the underlying characteristics of persuasive arguments. Perhaps the most relevant research to date is that in which subjects are asked to rate arguments or traits along dimensions such as how novel they are (Vinokur & Burnstein, 1974), or how many implications they elicit (Burnstein & Schul, 1982), in order to determine what aspects of information enhance its impact. Unfortunately, this important and rare work is in its infancy. After more than 40 years of research in experimental social psychology on persuasion, Fishbein & Ajzen (1981) could accurately state that "the general neglect of the information contained in a message... is probably the most serious problem in communication and persuasion research" (p. 359).¹

¹Notably, Fishbein and Ajzen (1975) and other expectancy-value theorists (e.g., Rosenberg, 1956) have examined argument or attribute persuasiveness from a phenomenological perspective. However, the question of *why* a particular argument or attribute is seen as more positive or negative than others is still not addressed.

In our own studies, we too have ignored the specific qualities that render some arguments cogent and others specious. Yet, for purposes of testing the ELM, it is necessary to specify arguments that people find compelling and those that are counterarguable. We have postponed the question of what specific qualities make arguments persuasive by defining argument quality in an empirical manner. In developing arguments for a topic we begin by generating a large number of arguments, both intuitively compelling and specious ones, in favor of some target issue (e.g., raising tuition). Then, members of the appropriate subject population are given these arguments to rate for persuasiveness. Based on these scores, we select arguments with high and low ratings to comprise at least one "strong" and one "weak" message. Subsequently, other subjects are given one of these messages and are told to think about and evaluate it carefully. Following examination of the message, subjects complete a "thought listing measure" (Brock, 1967; Greenwald, 1968) in which they are instructed to record the thoughts elicited by the message. These thoughts are then coded as to whether they are favorable, unfavorable, or neutral toward the position advocated (e.g., see Cacioppo & Petty, 1981c; Cacioppo, Harkins, & Petty, 1981).

We define a "strong" message as one containing arguments such that when subjects are *instructed* to think about the message, the thoughts that they generate are predominantly favorable. A "weak" message is one that is also ostensibly in favor of the advocacy. However, the arguments in a weak message are such that when subjects are instructed to think about them, the thoughts that they generate are predominantly unfavorable. A message that elicits a roughly even mixture of favorable and unfavorable thoughts would be considered an ambiguous or "mixed" message. Importantly, for positive attitude change to occur in response to a strong message, the profile of issue-relevant thoughts elicited by the arguments should be more favorable than the profile available prior to message exposure. For negative attitude change (boomerang) to occur in response to a weak message, the profile of thoughts elicited by the arguments should be less favorable than the profile available prior to message exposure.

Once the messages meet the criterion of eliciting the appropriate profile of thoughts, they are checked for other characteristics. First, a panel of subjects rates the messages for overall believability. For example, consider a message that argues that the University Chancellor wants to raise tuition so that more books can be purchased for the library. Students find this to be a strong and believable argument. Consider another message that argues that the Chancellor wants to raise tuition so that more trees and shrubs can be planted on campus. Students find this to be a weak but believable argument. Thus, although these arguments are equated in believability, they differ greatly in strength. It is, of course, possible to generate arguments that would appear even stronger or weaker but that would be completely implausible. For example, consider the argument that the Chancellor wants to raise tuition to build a 100-story monument to the university ping pong team.

Certainly a weak argument, but not one that would be taken seriously. Likewise, consider the argument that tuition should be raised because if it is, the university could improve enough in a year to become the best university in North America. Students would certainly find this consequence to be desirable, but because they wouldn't believe that it would occur, the argument would likely be derogated. In sum, we attempt to develop arguments that are strong and weak, but that do not strain credulity (Osgood & Tannenbaum, 1955). This is not to say that our arguments are necessarily veridical—just reasonably plausible to our subjects. The Appendix at the end of this chapter provides examples of strong and weak arguments that we have used to construct the messages used in some of our studies. Importantly, since the plausibility and strength of these arguments may vary widely across different populations and times, the arguments should be pretested and adapted as necessary before usage (cf., Cacioppo, Petty, & Morris, 1983).

Finally, a panel of subjects from the relevant subject pool rates our strong and weak messages for overall comprehensibility, complexity, and familiarity. Again, our goal is to develop strong and weak messages that are roughly equivalent in their novelty and in our subjects' ability to understand them.² The top panel of Figure 2-3 depicts the results of a study in which a treatment variable has no effect on attitudes. In the right panel, conditions of high elaboration likelihood are shown. In this hypothetical study, only the quality of the message arguments determined the extent of attitude change. In the left panel, conditions of low elaboration likelihood are depicted. In this hypothetical study, neither the treatment nor the arguments in the message affected attitudes. This result would occur when subjects have little motivation and/or ability to process the issue-relevant arguments, and Factor-X neither serves as a peripheral cue nor affects argument processing. In subsequent sections of this chapter we compare the two simple results shown in the top panel of Figure 2-3 with the other possibilities depicted in the figure.

Peripheral Cues

According to the Elaboration Likelihood Model, one way to influence attitudes is by varying the quality of the arguments in a persuasive message. Another possibility, however, is that a simple cue in the persuasion context affects attitudes in the absence of argument processing. As we noted earlier, some cues will do this because they trigger relatively primitive affective

²Although we do not detail the procedures here, it is important to note that after participation in one of our persuasion experiments, subjects receive a thorough debriefing as to the nature of the research and any deceptions employed and their rationale.

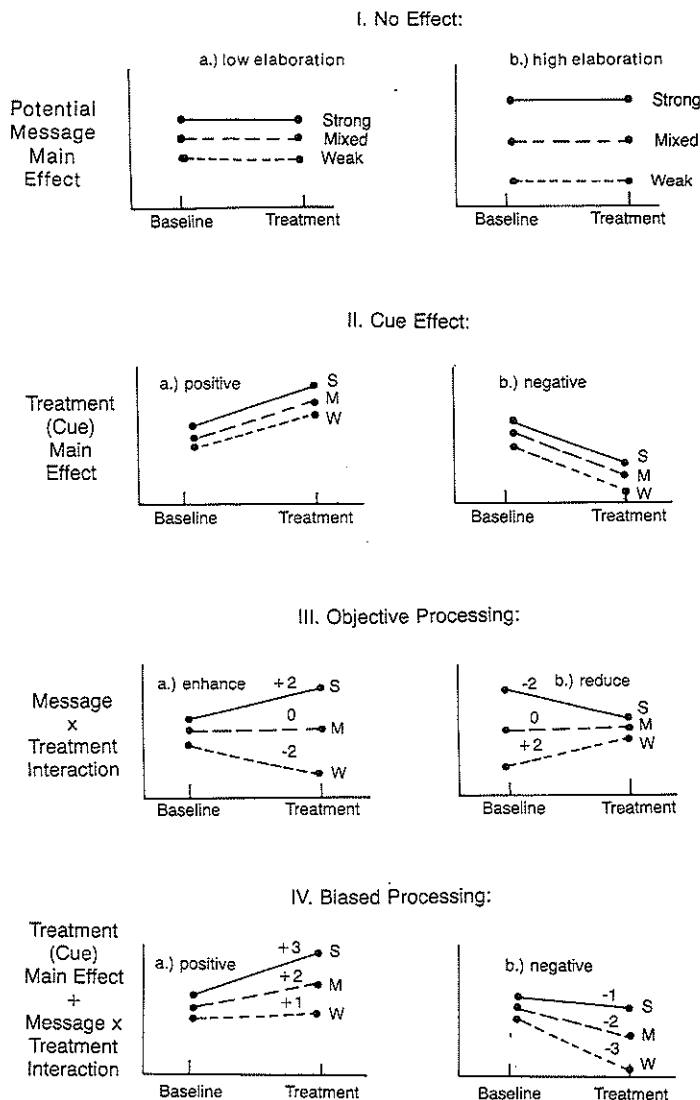


Figure 2-3. Possible effects of a treatment according to the ELM. In panel I the treatment has no effect. Instead, only an effect for argument quality is observed, at least if the elaboration likelihood is high. In Panel II, the treatment serves as a positive or negative cue affecting attitudes in the absence of much argument scrutiny. In Panel III, the treatment enhances or reduces argument scrutiny in a relatively objective manner. In Panel IV, the treatment produces either a positive or a negative bias to the message processing (adapted from Petty & Cacioppo, 1984c, 1986).

states that become associated with the attitude object. Other cues work, however, because they invoke guiding rules or inferences. (e.g., if I bought it, I must like it; Bem, 1972).

Since cues are postulated to affect attitude change without dramatically affecting argument processing, it is possible to test manipulations as potential cues by presenting them to subjects with the advocated position only (i.e., without accompanying persuasive arguments), as in prestige suggestion (see Asch, 1948). If the manipulation is a potential cue, it should have the ability to affect attitudes in the absence of any arguments. Alternatively, one could present an incomprehensible message (e.g., in a foreign language) on some topic along with the potential cue (e.g., speed of speech; Miller, Maruyama, Beaber, & Valone, 1976). Subjects could be asked to rate, for example, how likely it is that the speaker is convincing. Again, if a factor is operating as a cue it should be capable of affecting judgments even if there are no arguments to process. Finally, a simple procedure might involve merely describing various potential cues to subjects (e.g., a message with 1 vs. 10 arguments; a message from an attractive vs. an unattractive source) and asking them which would more likely be acceptable and/or persuasive. It is important to note that some cues will be difficult to uncover using the self-report procedure. For example, people may come to dislike a person more after an interaction in an uncomfortably hot than in a comfortably cool room, but may not be aware of the connection (Griffitt & Veitch, 1971). Conversely, the self-report procedure may uncover cues that people think would work, but in fact do not (Nisbett & Wilson, 1977). Furthermore, none of the procedures that we have suggested to uncover potential cues are capable of indicating *why* the cue was effective (e.g., affective association or the invocation of a simple decision rule). Nevertheless, these procedures should prove useful in suggesting whether a manipulation has the *potential* to serve as a peripheral cue.

The second panel in Figure 2-3 presents the results of a hypothetical study in which strong, weak, and mixed argument messages were presented along with a treatment that served as a peripheral cue. Note that in the pure case of cue processing, the cue affects all three kinds of messages equally. Since cues are most likely to operate when subjects are either unmotivated or unable to process issue-relevant arguments (as diagrammed in Figure 1-1), the data show a strong main effect for the cue treatment, but little effect for argument quality. In the left half of panel II in Figure 2-3 the cue is positive, and in the right half the cue is negative.

Message Elaboration

We have now described two of the key constructs in the Elaboration Likelihood Model: argument quality and peripheral cues. The third way in which a variable can affect persuasion is by determining the extent or

direction of message processing. As we explained in Chapter 1, variables can affect argument processing in either a relatively objective or in a relatively biased manner (Petty & Cacioppo, 1981a). Recall that by relatively *objective* processing, we mean that the variable either motivates or enables subjects to see the strengths of cogent arguments and the flaws in specious ones, or inhibits them from doing so. By relatively *biased* processing we mean that the variable either motivates or enables subjects to generate a particular kind of thought in response to a message, or inhibits a particular kind of thought. We discuss variables that affect relatively objective processing in Chapters 3 and 4 and we discuss variables that affect relatively biased processing in Chapter 5.

Of course, in order to test the ELM, it is important to assess how much message processing subjects are engaged in (i.e., how much cognitive activity or effort is devoted to issue-relevant thinking). We have employed five different procedures to assess the extent of elaboration in our research on persuasion: (1) self-reports of effort, (2) argument recall, (3) thought-listing, (4) electrophysiological activity, and (5) manipulation of argument quality. We discuss each of these briefly below.

Self-Reported Cognitive Effort

The simplest procedure to assess the extent of thinking involves asking people directly how much effort they expended in processing the message or how much thinking they were doing about the advocacy. For instance, subjects might use a 9-point scale to respond to the following questions: (a) "To what extent were you trying hard to evaluate the communication?"; and (b) "How much effort did you put into evaluating the communication?" We have found answers to these questions to be highly interrelated (e.g., average correlation > 0.80), so each subject's average response to scales like these might serve as an index of perceived cognitive effort (e.g., Petty, Harkins, Williams, & Latane, 1977). Alternatively, subjects might be asked to rate the extent to which they generated "many thoughts" or "few thoughts" about the message (Batra & Ray, 1986). Although we have found these methods sensitive in some studies (e.g., Cacioppo, Petty, & Morris, 1983; Petty, Harkins, & Williams, 1980), in others it has not produced differences even though there were other indications of differential processing (e.g., Harkins & Petty, 1981a, 1982). The problem, of course, is that although people may sometimes be aware of how much cognitive effort they are expending, people do not always have access to their cognitive processes (Nisbett & Wilson, 1977). In addition, the measures would prove invalid indications of issue-relevant thinking to the extent that responses reflected cognitive effort or thoughts devoted to processing peripheral features of the message (e.g., the color of paper it was printed on).

In a similar vein, subjects might be asked to rate the extent to which they were trying to generate favorable or unfavorable thoughts or the extent to which it was easy or difficult to generate a particular kind of thought in an

attempt to assess biased processing. Using this self-report procedure, Fukada (1986) found that subjects reported engaging in more counter-arguing during a message that followed a forewarning of persuasive intent than one that did not. This replicated the findings of previous research in which subjects actually listed their thoughts after a message with and without this type of forewarning (Petty & Cacioppo, 1979a; see further discussion in Chapter 5). Importantly, these measures suffer the same limitations as self-reports of relatively objective processing.

Argument Recall

A procedure stemming from the work on depth of processing (Craik & Lockhart, 1972) and cognitive effort (Tyler, Hertel, McCallum, & Ellis, 1979; see Chapter 1) is to give subjects five to ten minutes to list all of the arguments from the communication that they can remember. Two judges, blind to the experimental hypotheses, score the arguments recalled. An argument is counted as recalled if it correctly summarizes one of the arguments in the persuasive message. Listings of different versions of the same argument, or listings of different aspects or data used to support a given argument, are scored as a single argument correctly recalled. The interrater reliability is computed, and with few exceptions, extremely high agreement (i.e., $r_s > + 0.90$) is obtained.

Among the insensitivities one can anticipate using this measure are that: (a) free recall is an imperfect measure of encoding efficacy (Eagly & Chaiken, 1984), and (b) according to the ELM, individuals in different persuasion contexts (e.g., low vs. high distraction) may be equally able to report the message arguments presented but may differ greatly in the extent to which the presentation of the arguments caused them to think about issue-relevant information and experiences (e.g., Petty, Wells, & Brock, 1976). For instance, an individual who is able but unmotivated to think about a message may have effectively encoded four of five of the arguments presented without thinking much about the personal implications of any of these arguments; an individual who is both able and motivated to think about a message, on the other hand, may have effectively encoded and elaborated on four of the five arguments. A message recall measure taken shortly after communication exposure would be unlikely to distinguish between the message processing of these individuals. In our own research, although differences between high and low elaboration likelihood conditions have not generally produced significant differences in the number of message arguments recalled, high elaboration likelihood has tended to be associated with more argument recall than low elaboration likelihood.

Importantly, argument recall may also prove useful in indexing the extent of biased processing. Specifically, consider two subjects who are presented with five strong arguments on each side of a controversial issue. The relative number of pro to counterattitudinal arguments recalled may index the extent to which subjects were either particularly motivated or able to process

one side over another (cf., Jones & Aneshansel, 1956). Similarly, if a subject is presented with both strong and weak arguments in favor of his position, the extent to which strong arguments are recalled relative to weak ones would indicate bias. If the arguments are opposed to one's position, then biased processing would more likely favor recall of weak rather than strong arguments (cf., Jones & Kohler, 1958; Kleinhesselink & Edwards, 1975).

Thought-Listing Technique

A third procedure for assessing elaboration involves using the thought-listing technique pioneered by Brock (1967) and Greenwald (1968). In this procedure, subjects list their thoughts either in anticipation of, during, or after message exposure, and the thoughts are subsequently categorized into theoretically meaningful units (e.g., favorable and unfavorable thoughts; source-related thoughts) by the subjects or independent judges. For instance, subjects might read:

We are now interested in what you were thinking about during the last few minutes. You might have had ideas all favorable to the recommendation, all opposed, all irrelevant to the recommendation, or a mixture of the three. Any case is fine; simply list what it was that you were thinking during the last few minutes. The next page contains the form we have prepared for you to use to record your thoughts and ideas. Simply write down the first idea that comes to mind in the first box, the second idea in the second box, etc. Please put only one idea or thought in a box. You should try to record only those ideas that you were thinking during the last few minutes. Please state your thoughts and ideas as concisely as possible . . . a phrase is sufficient. Ignore spelling, grammar, and punctuation. You will have 2.5 minutes to write your thoughts. We have deliberately provided more space than we think most people will need to ensure that everyone would have plenty of room to write the ideas they had during the message. So don't worry if you don't fill every space. Just write down whatever your thoughts were during the last few minutes. Please be completely honest and list all of the thoughts that you had. (Petty & Cacioppo, 1977)

Following these instructions, subjects are provided with boxes (e.g., created by twelve 8-inch horizontal lines each about 1 inch from the one above) in which they are to write their ideas. Subsequently, these verbal protocols are content analyzed along some dimension(s), such as *polarity*, where each listed thought is rated as being in favor of the advocated position, opposed to the advocated position, or neutral/irrelevant. A more detailed discussion of the thought-listing technique, including issues regarding the eliciting instructions, listing interval, and scoring procedures, is provided in Cacioppo and Petty (1981c) and Cacioppo, Harkins, and Petty (1981). Suffice it to say that the thought-listing technique has proven to be an important supplemental tool in tracking the amount and type of cognitive activity involved in persuasion and resistance.

Two aspects of the listed thoughts may shed light on the extent and nature of information processing. First, if Factor-A increases the total number of issue- or message-relevant thoughts subjects list in a specified period of time

relative to Factor-B, then it may be reasonable to assume that Factor-A increases thinking (see Burnkrant & Howard, 1984; Brickner, Harkins & Ostrom, in press). *Issue-relevant* thoughts refer to all thoughts listed that relate to the topic of the advocacy. A subcategory of these thoughts includes those that were clearly sparked by or represent reactions to the specific message arguments presented (*message-relevant*). It might appear ideally that one would want to analyze only message-relevant thoughts to assess the extent of message processing, but this category likely underrepresents the extent of message processing. This is because many of the thoughts falling in the more global issue-relevant category may have been sparked by the message, although this cannot be definitively determined by examining the thought content. For this reason, even though it is possible to reliably code thoughts into issue- and message-relevant categories (e.g., Heesacker, Petty, & Cacioppo, 1983), it may sometimes be desirable to use the more global measure of issue-relevant thoughts to infer the extent of message processing.

Specifically, the relative *profile* of favorable and unfavorable issue-relevant thoughts listed may be used to index the extent of thinking even if there are no differences in the absolute number of thoughts listed. This follows from the fact that subjects are likely to list some issue-relevant thoughts during a postmessage thought-listing even if the specific message arguments elicited few thoughts during exposure. Importantly, issue-relevant thoughts that were not influenced by the message arguments should be guided primarily by the subject's initial attitude on the topic. Therefore, to the extent to which Factor-A elicits a profile of issue-relevant thoughts that better reflect the quality of the issue-relevant arguments presented than Factor-B, it can be assumed that Factor-A has enhanced argument processing relative to Factor-B. For example, consider a person who has not expended much effort processing a counterattitudinal message containing very strong arguments. When given 3 minutes to list thoughts about the message, this person might list 3 unfavorable and 2 favorable issue-relevant thoughts largely reflecting his or her original attitude. If the person had been given a message containing weak arguments, and the arguments were not processed, the profile of listed thoughts should be similar (i.e., 3 unfavorable, 2 favorable). Now consider a person with the same initial attitude who diligently processed a message containing cogent arguments in support of the advocacy. This person might list 4 favorable and only 1 unfavorable thought during the three-minute thought-listing task. If the strong arguments were processed, then the person should be *more* likely to list favorable thoughts but *less* likely to list unfavorable thoughts than if the message was not processed. The total number of issue-relevant thoughts might therefore be unaffected. Similarly, if this high-processing person had been exposed to a message containing weak arguments, then the profile of thoughts listed should be more likely to contain unfavorable thoughts and less likely to contain favorable thoughts than if the message was not processed. Note that both high and low processing led to the

production of 5 issue-relevant thoughts. However, in the low-processing conditions, the profile of thoughts is unaffected by the manipulation of argument quality (since the thoughts are guided by the person's initial attitude), whereas in the high-processing conditions, the thoughts are affected by argument quality. The more the profile of thoughts reflects the quality of arguments presented, the more likely it is that the arguments have been processed. Finally, it is important to note that although statistical procedures have been used to show that cognitive activity (as assessed by thought-listings) mediates attitude effects in some instances (e.g., Cacioppo & Petty, 1979b; Insko, Turnbull, & Yandell, 1974; Petty & Cacioppo, 1977), thought-listings alone do not provide definitive evidence for the cognitive mediation of attitudes because the evidence is basically correlational (cf., Miller & Colman, 1981).

Electrophysiological Responses

Cognitive activity. A fourth procedure that we have used to assess the extent and affectivity of information processing activity involves the use of psychophysiological measures. For example, we have completed several studies on perioral EMG activity and silent language processing (Cacioppo & Petty, 1979a; Cacioppo, Petty, & Marshall-Goodell, 1984; Cacioppo, Petty, & Morris, 1985). In an illustrative study, subjects participated in a 2 (Experimental replication) $\times$ 5 (Processing task) $\times$ 12 (Task replication) mixed-model factorial design, with the first factor manipulated between-subjects (Cacioppo & Petty, 1981b). Subjects were exposed to 60 trait adjectives spanning a range of likability, and each trait adjective was preceded by one of five cue-questions defining the processing task. The cue-questions were: (a) "Is the following word spoken louder than this question?" (Volume discrimination), (b) "Does the following word rhyme with——?" (Rhyme), (c) "Is the following word similar in meaning to——?" (Association), (d) "Is the following word good (bad)?" (Evaluation), and (e) "Is the following word self-descriptive?" (Self-reference). Results revealed that mean recognition confidence ratings were ordered (from highest to lowest) as follows: self-reference, evaluation, association, rhyme, and volume discrimination. Importantly, all means except the last two differed significantly from each other (see left panel, Figure 2-4).

Analyses of EMG activity in this study revealed that the mean amplitude of the EMG activity during the task over the perioral muscle region (*orbicularis oris*—which controls the pursing action of the lips) was lowest for the tasks requiring minimal associative processing (volume discrimination, rhyme), intermediate for the task of simple association, and equally high for the tasks of evaluation and self-reference. This pattern of data (see middle panel in figure 2-4) is unlikely to be due to task-induced differences in physiological arousal, because further analyses revealed that: (a) cardiac activity and the mean amplitude of EMG activity over a nonoral muscle group (i.e., the nonpreferred forearm flexors region) did not vary as a

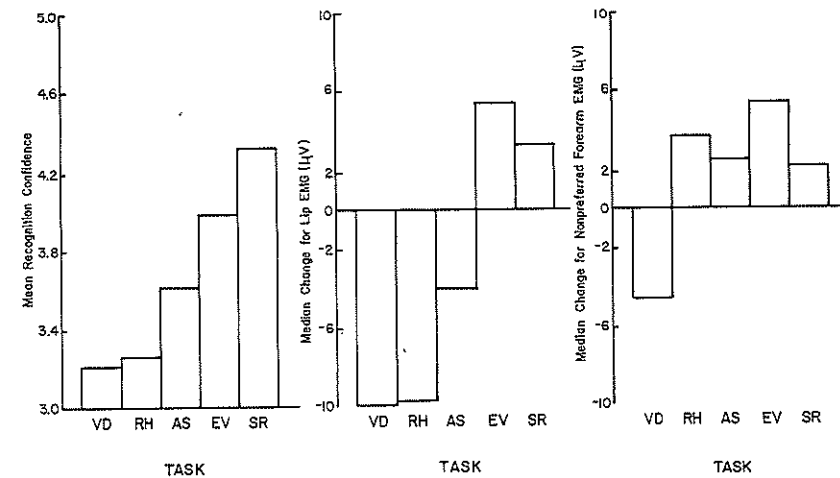


Figure 2-4. Psychophysiological assessment of extent of processing. The figure depicts the mean recognition confidence (left panel) and the median change from prestimulus levels for electromyographic activity during the processing interval for lip (middle panel) and nonpreferred forearm (right panel) as a function of orienting task. Abbreviations: VD = volume discrimination; RH = rhyme; AS = association; EV = evaluation; SR = self-reference (see text for specific orienting questions) (adapted from Cacioppo & Petty, 1981b).

function of the type of task performed (see right panel in Figure 2-4); and (b) the association between task and EMG activity over the perioral region was temporally as well as spatially specific. In other words, task-discriminating EMG activity was observed only during the period in which subjects analyzed the aurally presented trait adjectives and formulated their responses. Finally, in a follow-up study, we found that different associative tasks, such as evaluative and self-referent tasks, have discriminable effects on perioral EMG activity, although these effects are oftentimes on the form rather than the mean amplitude of the task-evoked response (Cacioppo, Petty, & Morris, 1985).

Affective reactions. Although the muscles of the lower face and perioral region are clearly involved in expressions of emotions (e.g., see Ekman & Friesen, 1978), perioral EMG activity does not appear to be related to subtle variations in the affective tone of individuals' information processing, such as those typically involved in attitude studies. For instance, some of the trait words employed in the Cacioppo and Petty (1981b) study were positive, others were neutral, and others were negative in meaning. We found no differences in perioral EMG activity as a function of word-valence, even though we have observed differential hemispheric EEG activity in response to these stimuli (Cacioppo & Petty, 1980c).

While the extent of cognitive deliberation rather than people's affective reactions has been discernible in perioral EMG activity, EMG activity over muscles controlling the movement of facial landmarks in regions ranging from the lower face (e.g., *zygomatic major*, which pulls the corners of the mouth upwards and back when forming a smile) to the upper face (e.g., *corrugator supercilii*, which draws the brows together and down in negative emotions such as anger and sadness) has been found to vary as a function of the nature and intensity of affective reactions to stimuli and the affective tone of attitudinal processing (e.g., Cacioppo & Petty, 1979a; McHugo, Lanzetta, Sullivan, Masters, & Englis, 1985; see Figure 2-5 for electrode placements). In one illustrative study, subjects were led to believe they were participating in a study on involuntary neural responses during action and imagery (Cacioppo, Petty, & Marshall-Goodell, 1984). Subjects on any given trial either lifted (action) or imagined lifting (imagery) a "light" (16 g) or "heavy" (35 g) weight, or they either silently read (action) a neutral communication as if they agreed or disagreed with its thesis, or imagined reading (imagery) an editorial with which they agreed or disagreed. Results revealed that EMG activity over the perioral region was higher during the attitudinal tasks (which involved silent language processing) than physical tasks, but perioral EMG activity was again essentially unchanged by the affective tone of the subjects' attitudinal processing. Yet whether subjects thought about the communication in a positive or negative fashion influenced EMG activity over facial muscles of emotional expression. For instance, positive attitudinal processing was associated with lower EMG

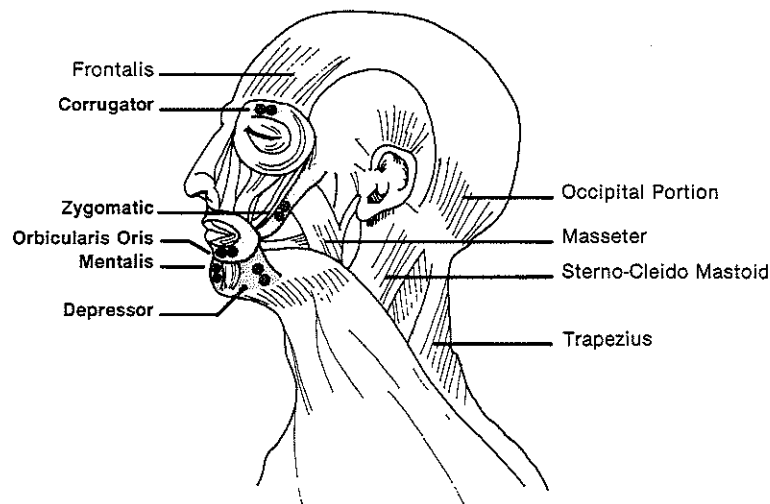


Figure 2-5. Electrode placements used for assessing extent and affectivity of processing (adapted from Cacioppo & Petty, 1981b).

activity over the corrugator region and higher EMG activity over the zygomatic region than was negative attitudinal processing. Finally, EMG activity recorded over the superficial forearm flexors did not vary as a function of the attitudinal tasks employed in this study, but it (rather than facial EMG activity) varied as a function of the simple physical tasks.

In another study (Cacioppo, Petty, Losch, & Kim, 1986), we had subjects view slides of scenes (e.g., a mountain cliff) that were mildly to moderately evocative of positive or negative affect. Pilot testing revealed that although there was consistency across individuals in their ratings of the affective valence of the scenes, there was considerable variation among individuals in terms of the rated intensity of the affect evoked by a given slide. Hence, we used the apriori classifications of *valence* as determined in pilot testing, but employed subjects' own ratings of the stimuli to determine *intensity* of the slides (i.e., which stimuli were viewed as mildly or moderately positive or negative). Electrodes were placed at appropriate locations on a subject's face (see Figure 2-5), and additional dummy electrodes were attached to the head and torso to divert attention from the face as a particular site of interest. After a period of relaxation, subjects were exposed to 16 habituation slides and then to another 32 during which data were collected. Each slide was projected for 5 seconds and after each presentation subjects used a numeric keypad to answer questions presented on a video screen before them. Subjects' faces were videotaped as they responded to the slides and subsequent analyses revealed that independent judges, blind to condition, were unable to determine from the tapes whether a positive or negative stimulus had been presented or whether a mildly or moderately intense stimulus had been viewed. However, EMG activity over the corrugator, zygomatic, and orbicularis oculi regions differentiated the pleasantness and intensity of individuals' affective reactions to the slides. These results suggest that gradients of EMG activity over the muscles of the face can provide objective and continuous probes of affective processes that are too subtle or fleeting to evoke expressions observable under normal conditions of social interaction (see Cacioppo & Petty, in press for further discussion).

Objective vs. biased processing. We have reviewed some evidence that electromyographic activity may be used to assess the extent and affectivity of information processing. It may also be possible to index the extent to which information processing activity is relatively objective rather than biased by assessing the extent of differential hemispheric activation during message processing. For example, in two studies we found that individuals who produced a relatively polarized profile of thoughts (unfavorable-favorable thoughts) in response to a persuasive message also demonstrated relative right hemispheric EEG activation as assessed over the parietal lobes (Cacioppo, Petty, & Quintanar, 1982; see further discussion in Chapter 5). If the EEG ratio is used in combination with measures of facial EMG, it might

be possible in some situations to index both the extent and direction of elaboration. For example, a relatively high score on perioral EMG might indicate that a person was doing a lot of thinking about an issue, and a pattern of relative right hemispheric activation might indicate that this thinking was relatively one sided. Corrugator and zygomatic activity might then be used to determine the affective tone of the processing.

Summary. The physiological procedures we have described in this section have several potential advantages over self-reports of cognitive activity, argument recall, and thought-listings. For example, these measures can track psychological processes over time and may be less susceptible to artifacts (e.g., demand characteristics) and subjects' inability to recall the process or content of their thoughts. Although work on psychophysiological assessments of attitudinal processes is in its early stages, it holds considerable promise for tracking and marking the underlying mediation of persuasion and resistance (for further information, see Cacioppo & Petty, 1981a, in press; Petty & Cacioppo, 1983b).

Message/Argument Quality

The final procedure for assessing the extent of cognitive processing and the one we highlight in this monograph is based on the manipulation of message argument quality described earlier in this chapter. As we hinted above, our empirical method of defining argument quality allows us to assess the extent to which a variable affects argument processing and the extent to which this processing is relatively objective or relatively biased. We'll consider first the expected consequences of variables affecting relatively objective processing.

Relatively objective processing. Assume for the moment that we have created a control condition in which motivation or ability to process issue-relevant arguments is rather low. Subjects should show relatively little differentiation of strong from weak arguments in this condition. However, if a manipulation enhances argument processing in a relatively objective manner, then subjects should show greater differentiation of strong from weak arguments. More specifically, a message with strong arguments should tend to produce more agreement when it is scrutinized carefully than when scrutiny is low, but a message with weak arguments should tend to produce less overall agreement when scrutiny is high rather than low. The joint operation of these processes would result in people showing greater attitudinal differentiation of strong from weak arguments when processing is high than when it is low. This pattern of results is depicted in the left half of panel III in Figure 2-3. In a similar fashion we can assess the extent to which a variable disrupts processing in a relatively objective manner. Consider a situation in which subjects are processing the message arguments quite diligently. These subjects should show considerable differentiation of strong from weak arguments. However, if argument processing is

disrupted, either because of reduced motivation or ability, argument quality should be a less important determinant of persuasion. More specifically, a strong message should tend to induce less agreement when processing is disrupted than when it is not, but a weak message should tend to produce more agreement when processing is disrupted than when it is not. The right half of panel 3 in Figure 2-3 depicts this pattern. In addition to subjects' attitudes being more differentiated to weak and strong messages when argument processing is high rather than low, as we noted earlier in this chapter, subjects' thoughts should also show greater differentiation of arguments when processing is high rather than low. Importantly, our predictions about discriminating arguments are *relative* rather than absolute. As processing increases, attitudes are more responsive to manipulations of argument quality, and as processing decreases, attitudes are less responsive to manipulations of argument quality.

Relatively biased processing. Panel IV in Figure 2-3 graphs the expected results for a variable that biases information-processing activity. In the left half of the panel, the effects of a variable that produces a positive cognitive bias (enhancing favorable thoughts and/or reducing negative thoughts) is depicted. It is instructive to compare this pattern of data with the patterns in the two panels above it. First, note that unlike a variable operating as a simple positive cue (left half of panel II), a variable producing a positive processing bias is not expected to affect all messages equally. Since the pure cue processor is not elaborating message arguments at all, the effectiveness of the cue is not constrained by the arguments presented. The biased processor, however, is attempting to elaborate the arguments and in this regard is similar to the objective processor. Nevertheless, an important difference between relatively objective and biased processing exists. The objective processor is motivated or is able to discover the "true (subjective) validity" of the message, and thus, strong arguments induce more persuasion and weak arguments induce less persuasion with more processing. In stark contrast, the biased processor is either particularly motivated or able to generate a particular kind of thought, often in defense of an initial attitude. However, even though the person is biased in processing a communication, the arguments in the message pose some limitation on this bias. For example, consider a person who is truly motivated to *counterargue* (and not simply discount) an advocacy. This person's task is simpler to the extent that the message provides weak rather than strong arguments in support of its position (see right half of panel IV in Figure 2-3).

As panel IV indicates, a treatment that biases thinking in a positive direction should generally have a greater impact on a strong than a weak message because it will be more difficult for a person to generate thoughts favorable to weak than to strong message arguments. On the other hand, a variable that biases thinking in a negative direction should generally have a greater impact on a weak than a strong message because it will be more

difficult for a person to generate counterarguments to strong than weak arguments.

Importantly, these predictions (and the depictions in panel IV of Figure 2-3) assume that in the baseline (control) condition, relatively little issue-relevant thinking is occurring. However, consider a control (comparison) condition in which subjects are maximally processing strong and weak arguments. If the experimental treatment includes a variable that biases thinking in a positive direction, it will be difficult to observe more favorable attitudes to the strong arguments in the experimental than the control condition inasmuch as the arguments are already being processed maximally in the control cell (i.e., a ceiling effect is operating). However, the positive bias may result in more favorable attitudes toward the weak message than observed in the control condition (because no ceiling effect is operating). Thus, it may appear that the positive bias is working better for the weak than the strong message.

Similarly, if the experimental treatment includes a variable that biases thinking in a negative direction, it will be difficult to observe more negative attitudes toward the weak arguments than in the control condition if control subjects are highly motivated and able to process the message objectively (without bias). Thus, it may appear that the negative bias is working better for the strong than the weak message (because of a floor effect for the weak arguments). The caveat here is to include an appropriate control or baseline condition so that floor and ceiling effects do not cause problems. In general, when testing variables hypothesized to enhance processing it is better to include control conditions in which processing is minimal. When testing variables hypothesized to reduce processing, the opposite holds.

Self-reports of message quality. It is now clear that if a variable enhances objective or biased processing, subjects' attitudes should better discriminate strong from weak arguments when that variable is present. If the variable disrupts processing, subjects' attitudes should more poorly discriminate strong from weak arguments when that variable is present. A spin-off from the preceding technique that we have employed to assess the extent to which subjects carefully considered the cogency of the message arguments is to have subjects provide an overall evaluation of message quality. For instance, subjects might respond to the following questions: (a) "To what extent do you feel the communication made its point effectively?" (1 = not at all, 9 = completely), (b) "To what extent do you feel the communication was convincing?" (1 = not at all, 9 = completely). A subject's average response to these questions would serve as a measure of message evaluation.

As was the case when interpreting the effects of argument quality on postcommunication attitudes, the interesting outcome is not whether strong arguments are rated as being of higher quality than weak arguments, but rather how some variable combines with the manipulation of argument quality to affect subjects' overall evaluation of the cogency of the message.

For instance, if varying the level of some variable increases the extent to which recipients scrutinize the externally provided message arguments, then: (a) subjects' overall evaluations of strong versus weak versions of the message arguments should be more discriminable, and (b) because the postcommunication attitudes are more likely to reflect the thoughts and ideas evoked by the persuasive communication, the correlation between message evaluation and postcommunication attitudes should be higher.

Summary. Figure 2-3 summarizes the ways in which a treatment can affect attitude change according to the ELM, and it shows how these different processes can be tested by varying argument quality. First, a treatment can have no effect on persuasion for either strong or weak arguments (such as a peripheral cue under conditions of high elaboration likelihood; right side of panel I). Second, a treatment may produce only a main effect enhancing or reducing persuasion regardless of argument quality (panel II). If so, it suggests that the treatment is operating as a simple positive or negative cue. However, if a treatment interacts with message quality, it suggests that the treatment is affecting the elaboration likelihood. If the interaction follows the form depicted in panel III of Figure 2-3, it suggests that effect of the variable on information processing is relatively objective. If the interaction follows the form depicted in panel IV of Figure 2-3, on the other hand, it suggests that the effect of the variable on information processing is relatively biased.

Need for Cognition: Assessing Elaboration Likelihood via Individual Differences

In presenting the ELM in Chapter 1, we have emphasized that people have neither the resources to think about every persuasive appeal nor the luxury (or apparently the inclination) of being able to ignore them all. In the preceding sections of this chapter we have suggested various means of assessing the elaboration likelihood either during or following the presentation of a persuasive communication. However, the fact that the situational factors that have been employed to manipulate the extent to which attitudes are based on issue-relevant thinking sometimes account for only a small portion of variance is theoretically due in part to systematic individual differences among people in their desire to engage in issue-relevant thinking when dealing with their social environment generally and when formulating their attitudes specifically. This suggested to us that an individual differences strategy could be employed to measure chronic differences among people in elaboration likelihood prior to the presentation of a persuasive appeal (Cacioppo & Petty, 1982, 1984b; Cacioppo, Petty, & Morris, 1983). In particular, we reasoned that there were stable (though not invariant) individual differences in intrinsic motivation to engage in effortful cognitive endeavors generally, just as there are stable individual

differences in intrinsic motivation to engage in effortful physical endeavors. Individual difference factors such as effectance, cognitive style, competency, and self-efficacy, although related when dealing with cognitive tasks to the theoretical distinction for which we were searching, failed to fully capture the specificity we sought. For instance, White (1959) described effectance motivation as follows:

...it is maintained that competence cannot be fully acquired simply through behavior instigated by (primary) drives. It receives substantial contributions from activities which, though playful and exploratory in character, at the same time show direction, selectivity, and persistence in interacting with the environment. Such activities in the ultimate service of competence must therefore be conceived to be motivated in their own right. It is proposed to designate this motivation by the term effectance, and to characterize the experience produced as a *feeling of efficacy*. (p. 329)

General differences across individuals in self-efficacy or effectance, therefore, should influence people's behavior on noncognitive (e.g., physical) as well as cognitive tasks.

The Construct of Need for Cognition

Our analysis of Cohen and his colleagues' work on *need for cognition* (1957; Cohen, Stotland, & Wolfe, 1955) suggested that it might be possible to scale individuals along their tendency to engage in and enjoy effortful cognitive endeavors. Cohen, Stotland, and Wolfe (1955) conceptualized need for cognition (NC) as "a need to experience an integrated and meaningful world." However, while one might avoid ambiguity and achieve an integrated and meaningful world through carefully scrutinizing incoming information, one might also achieve this goal by employing heuristics and by seeking the advice of experts. That this may be a feature of Cohen and his colleagues' conceptualization of NC is evident in the following description by Adams (1959):

Given the existence of a measurable need for cognition, as defined (by Cohen and his colleagues), it may be hypothesized that a mother whose child presents a behavioral problem which she cannot handle or about which she is uncertain will experience tension and will try to reduce the tension if she has a need for cognition. One way of reducing the attendant tension under these circumstances is to seek information or advice about the problem from an appropriate source. An appropriate source is seen as one which, in the past, under similar circumstances has reduced tensions resulting from frustration of the need for cognition. (p. 17)

It is also clear from the early research that the emphasis in Cohen and his colleagues' conceptualization of NC was on tension reduction (Cohen, Stotland, & Wolfe, 1955, p. 291). In contrast, the emphasis in contemporary research on NC is on the statistical tendency of and intrinsic enjoyment individuals derive from engaging in effortful cognitive activities (Cacioppo & Petty, 1982). In this sense, our conceptualization of NC embraces White's (1959) central thesis that there are directed and persistent behaviors that

have a motivational aspect which cannot be wholly derived from sources of energy conceptualized as drives or instincts. Individuals low in need for cognition (LNC) are conceived as being cognitive misers relative to individuals high in need for cognition (HNC), and this difference is conceived as attained slowly through repeated or prolonged episodes of effortful problem solving. When developing a method of measuring this individual difference construct (Cacioppo & Petty, 1982), we retained the term "need for cognition" despite our modifications to the original conceptualization in deference to Cohen's (1957) pioneering emphasis in persuasion research on individual differences in cognitive motivation (cf., Cohen, 1957).³

Scaling Individual Differences in Need for Cognition

Our initial efforts were aimed at constructing and validating an assessment instrument that tapped individual differences in people's intrinsic motivation to engage in and enjoy effortful cognitive endeavors (Cacioppo & Petty, 1982; Cacioppo, Petty, & Kao, 1984). In our initial study, we generated a pool of questions concerning a person's reactions to demands for effortful thinking in a variety of situations (Cacioppo & Petty, 1982, Experiment 1). When generating these, we specifically included items describing a variety of situations in which people could choose to garner information, analyze available evidence, abstract from past experiences, or synthesize ideas. We explicitly excluded items dealing with inner broodings, reverie, mystical or religious experience, mind wandering, and artistic imaginings. Hence, the need for cognition scale (NCS) was designed to distinguish between individuals who are intrinsically motivated to engage in and enjoy effortful analytic activities and those who are not. Following pilot testing to identify ambiguous items in need of rewording or deletion, the set of items was

³The original objective tests used by Cohen and colleagues to gauge individual differences in need for cognition were never described in detail and are apparently no longer available. It might be noted, however, that Rosen and his colleagues (Rydell & Rosen, 1966; Rosen, 1963, 1964; Rosen, Siegelman, & Teeter, 1964) also scaled an individual difference they termed the need for cognition, but they conceived of NC as representing cognitive motivation, broadly defined. Accordingly, their scale tapped twelve separate factors (e.g., "cognitive bookworm," "religious anti-intellectualism"), whereas like Cohen and colleagues, we sought to identify and measure a more specific and limited individual difference in cognitive motivation. It might also be noted that Rosen and colleagues never marshalled strong evidence for the stability and validity of their various scales. For instance, Rosen (1964) reported that students from a college honors program scored higher than liberal arts and sciences students on subscales such as cognitive-bookworm and social and religious anti-intellectualism, but the liberal arts and sciences students did not differ from, or actually scored lower than, students in a two-year general program on these subscales.

administered to two groups of individuals presumed to differ substantially in their tendency to engage in and enjoy effortful cognitive endeavors (i.e., members of the University of Iowa faculty vs. assembly line workers in the neighboring communities). Criteria of ambiguity, irrelevance, and internal consistency were employed to select the items for the NCS, and a factor analysis of these items confirmed that one factor was dominant. A second study, in which the NCS was administered to a more homogeneous population (undergraduate students in introductory psychology classes at the University of Missouri) again revealed one factor was dominant (Cacioppo & Petty, 1982, Experiment 2). Moreover, despite the dramatically different subject populations and testing circumstances, the factor loadings obtained in Studies 1 and 2 were found to be highly correlated ($r = +0.72$). Hence, the factor structure obtained in the first study was cross-validated. To simplify its administration, we subsequently developed a short-form of the NCS. This 18-item version of the NCS was found to possess the same factor structure as our original NCS and, indeed, correlated very highly ($r = +0.98$) with the original (Cacioppo, Petty, & Kao, 1984; see Table 2-1 for the specific items).

Additional evidence for the stability of the factor structure and internal consistency of the NCS was provided in three recent laboratory studies involving college undergraduates (Chaiken, in press) and in a field study involving a random sample of 233 residents from Gainesville, Florida (Ferguson, Chung, & Weigold, 1985). In the former studies, factor analyses revealed one dominant factor best characterized as a tendency to engage in and enjoy effortful cognitive endeavors. Ferguson et al. did not report a factor analysis, but rather they reported a Chronbach's alpha of 0.86 for the subset of 15 items they selected from the Cacioppo and Petty (1982) NCS.

Validation of the Need for Cognition Construct

It is important to know whether individuals high in need for cognition enjoy relatively effortful cognitive tasks even in the absence of feedback about performance. To examine this question, subjects were given either simple or complex instructions to use while performing a boring number-circling task (Cacioppo & Petty, 1982, Experiment 4). Subjects in the *simple* number-circling task condition were instructed to circle all 1s, 5s, and 7s. Subjects in the *complex* number-circling task condition were instructed to circle all the 3s, any 6 that preceded a 7, and every other 4. Subjects were given tables containing 3,500 random numbers, and all subjects performed the task for 10 minutes. Afterward, subjects expressed their attitude toward the task and completed the NCS, which was embedded among ancillary measures. Results revealed that subjects generally disliked the task, a not too surprising finding given its boring and tedious nature. More importantly, a significant interaction revealed that people high in need for cognition tended to prefer the complex to the simple task, whereas individuals low in need for cognition tended to prefer the simple to the complex task.

There is also evidence for the convergent and discriminant validity of our

Table 2-1. 18-Item Need for Cognition Scale

Item Number	Item Wording
1	I would prefer complex to simple problems.
2	I like to have the responsibility of handling a situation that requires a lot of thinking.
3	Thinking is not my idea of fun.*
4	I would rather do something that requires little thought than something that is sure to challenge my thinking abilities.*
5	I try to anticipate and avoid situations where there is likely chance I will have to think in depth about something.*
6	I find satisfaction in deliberating hard and for long hours.
7	I only think as hard as I have to.*
8	I prefer to think about small, daily projects to long-term ones.*
9	I like tasks that require little thought once I've learned them.*
10	The idea of relying on thought to make my way to the top appeals to me.
11	I really enjoy a task that involves coming up with new solutions to problems.
12	Learning new ways to think doesn't excite me very much.*
13	I prefer my life to be filled with puzzles that I must solve.
14	The notion of thinking abstractly is appealing to me.
15	I would prefer a task that is intellectual, difficult, and important to one that is somewhat important but does not require much thought.
16	I feel relief rather than satisfaction after completing a task that required a lot of mental effort.*
17	It's enough for me that something gets the job done; I don't care how or why it works.*
18	I usually end up deliberating about issues even when they do not affect me personally.

*Reverse scoring is used on this item.

Note: Subjects are asked to respond to the items on scales indicating their "agreement" or "disagreement," or to rate the extent to which the statements are "characteristic" or "uncharacteristic" of them (see Cacioppo & Petty, 1982).

conceptualization of need for cognition. In studies of personality variables we have found need for cognition to be: (a) positively and weakly related to field independence ($r = +0.19$, $N = 419$, $p < 0.05$), (b) negatively related to closed-mindedness ($r = -0.27$, $N = 104$, and $r = -0.23$, $N = 97$, $ps < 0.05$), (c) unrelated to Sarason's (1972) measure of test anxiety ($r = +0.02$, $N = 419$, n.s.), and (d) weakly if at all related to social desirability ($r = +0.08$, $N = 104$,

n.s., and $r = +0.21$, $N = 97$, $p < 0.05$). Olson, Camp, and Fuller (1984) found that the NCS correlated significantly with eight measures of curiosity (mean correlation = $+0.57$), but did not correlate significantly with social desirability, gender, or state or trait anger. Heppner, Reeder, and Larson (1983) found that subjects who scored high on a problem-solving inventory also had significantly higher NCS scores than subjects who scored low on the problem-solving inventory. Moreover, subjects high, in contrast to low, in need for cognition have been found to report possessing more prior knowledge about and confidence in their position on a broad range of attitude issues, and to express significantly more favorable attitudes toward positions that involved effortful thinking (e.g., instituting senior comprehensive exams, reading novels) but similar attitudes toward nonintellectual issues (e.g., pets, sports, university tuition, money) (Sidera, 1983; Cacioppo, Petty, & Morris, 1983, pp. 810-811).

Cacioppo and Petty (1984b) found further evidence of the specificity of need for cognition. Subjects completed an inventory identifying situations that aroused fear and anxiety. Seven broad categories of stressors were examined. Analyses revealed that HNC people expressed significantly less fear and anxiety about matters involving academic challenges (e.g., "evaluation of performance," "tests and grades") than LNC individuals, but similar levels of fear and anxiety about being harmed physically, observing others being harmed physically, dating, establishing or maintaining relationships, and nonacademic aspects of college life. Similar levels of overall fear and anxiety were also found for individuals high versus low in need for cognition.

Importantly, evidence has also shown that individuals high in need for cognition are more intrinsically motivated to engage specifically in effortful cognitive analyses. Recent research has demonstrated that individuals put less effort into a task when they share responsibility for the outcome as part of a group than when they are individually responsible for the outcome (e.g., Ingham, Levinger, Graves, & Peckham, 1974). This effect, which has been dubbed "social loafing" (Latane, Williams, & Harkins, 1979), is equally evident in cognitive (Harkins & Petty, 1982), attitudinal (Petty, Harkins, & Williams, 1980), and physical tasks (Williams, Harkins, & Latane, 1981). We reasoned that if HNC individuals are more intrinsically motivated to engage in effortful cognitive endeavors, then they should be less likely to socially loaf on an interesting cognitive task than LNC individuals (Petty, Cacioppo, & Kasmer, 1985). To test this hypothesis, we asked subjects to perform a brainstorming task (generating uses for objects) after they were led to believe that they were individually responsible or part of a group that was responsible for performing the task. Results revealed a significant interaction showing that LNC subjects generated fewer ideas under group than individual conditions, whereas HNC subjects generated equally high numbers of ideas regardless of social condition. For comparison purposes,

another group of subjects performed a cognitively unengaging physical task (screwing and unscrewing bolts and nuts) under individual or group instructions. Results revealed a main effect for social condition, showing greater loafing by subjects both low and high in need for cognition under the group conditions. These results are clearly consistent with the notion that individuals high in need for cognition are intrinsically motivated to engage in effortful cognitive endeavors in particular, and that this motivation has socially relevant behavioral consequences.

Although individuals high in need for cognition tend to engage in more effortful cognitive endeavors, this is not to suggest that their cognitive work is unfocused. In a study of person memory demonstrating this point, Srull, Lichtenstein, and Rothbart (1985) exposed subjects to congruent, incongruent, and irrelevant trait information. Results revealed that individuals high in need for cognition recalled more items than individuals low in need for cognition. This difference was most evident for incongruent items, moderate for congruent items, and nonsignificant for the items irrelevant to their forming a coherent impression. These data are consistent with the view that not only do individuals high in need for cognition think more about incoming information, but more specifically that they tend to elaborate upon previously unintegrated task-relevant information.

Finally, evidence for the notion that need for cognition can be used to assess chronic differences in elaboration likelihood via an individual differences approach has been obtained in field surveys as well as in the laboratory studies reviewed above. In a study conducted during the presidential election in the fall of 1984, Cacioppo, Petty, Kao, and Rodriguez (in press) matched subjects high and low in NC in terms of the attitudes they expressed toward the Democratic (Mondale/Ferraro) and Republican (Reagan/Bush) nominees 8 weeks prior to the election. Although this study is described in more detail in Chapter 7, for now we note that HNC subjects not only reported having thought more about and knowing more about the presidential candidates, but HNC individuals also listed significantly more facts about the presidential candidates than LNC individuals. Similarly, Ahlring and McClure (1985) found that significantly more HNC individuals reported intending to watch the presidential and vice-presidential debates and that HNC individuals actually produced significantly more thoughts regarding the consequences of electing a particular candidate than did LNC individuals. In another relevant study, Pieters (1986) surveyed residents of Columbia, Missouri to assess beliefs, attitudes, and intentions toward a city sponsored energy conservation program. The survey included an abbreviated version of the need for cognition scale. As expected, regression analyses revealed that the attitudes of high need for cognition subjects could be explained significantly by the perceived advantages and disadvantages of the energy program ($R = .62$), but this was not the case for those low in need for cognition ($R = .32$).

Summary

In sum, one of the major sources of variability in studies of attitude change is that attributable to individual differences. The ELM allows individual differences to be an integral part of theoretical thinking. Specifically, theory and research on need for cognition suggest that, unless overbearing situational constraints are operative, individuals high in need for cognition are characterized generally by higher levels of elaboration likelihood than individuals low in need for cognition. Hence, individuals high in need for cognition are consistently more likely to base their attitudes on a diligent analysis of relevant information, whereas individuals low in need for cognition should be more likely to utilize cognitively less taxing peripheral processes (see Cacioppo & Petty, 1984b; Heesacker, in press, for reviews). The role of need for cognition in persuasion is addressed explicitly in Chapters 4 and 6.

Retrospective

In this chapter, we discussed methodological factors involved in testing the Elaboration Likelihood Model of persuasion. After discussing our typical procedures for assessing attitudes and persuasion, we operationalized the key constructs in the ELM. Specifically, procedures for assessing the constructs of argument quality, peripheral cues, and message elaboration were addressed. We ended the chapter with a discussion of individual differences in "need for cognition" and the role of this construct in testing the ELM. In the remaining chapters of this book we present the empirical research assessing the validity of the ELM. This research employs the methodological features outlined in this chapter.

Appendix: Examples of Strong and Weak Arguments Used to Create Strong and Weak Messages in Favor of Implementing Senior Comprehensive Exams⁴

Example Strong Arguments

S1. The National Scholarship Achievement Board recently revealed the results of a five-year study conducted on the effectiveness of comprehensive

⁴These arguments, typical of those used in our research employing the senior comprehensive exam topic (e.g., Cacioppo, Petty, & Morris, 1983; Petty & Cacioppo, 1979b, 1984a) are reprinted for illustrative purposes. It is important to note that our conceptualization of *strong* and *weak* messages is tied not to particular statements or arguments (such as those above), but to an empirical criterion. Specifically, strong messages should elicit a profile of predominately favorable thoughts (e.g., 65%

exams at Duke University. The results of the study showed that since the comprehensive exam has been introduced at Duke, the grade point average of undergraduates has increased by 31%. At comparable schools without the exams, grades increased by only 8% over the same period. The prospect of a comprehensive exam clearly seems to be effective in challenging students to work harder and faculty to teach more effectively. It is likely that the benefits observed at Duke University could also be observed at other universities that adopt the exam policy.

S2. Graduate schools and law and medical schools are beginning to show clear and significant preferences for students who received their undergraduate degrees from institutions with comprehensive exams. As the Dean of the Harvard Business School said: "Although Harvard has not and will not discriminate on the basis of race or sex, we do show a strong preference for applicants who have demonstrated their expertise in an area of study by passing a comprehensive exam at the undergraduate level." Admissions officers of law, medical, and graduate schools have also endorsed the comprehensive exam policy and indicated that students at schools without the exams would be at a significant disadvantage in the very near future. Thus, the institution of comprehensive exams will be an aid to those who seek admission to graduate and professional schools after graduation.

S3. A member of the Board of Curators has stated publicly that alumni nationwide have refused to increase their contributions to the University because of what they feel are lax educational standards. In fact, the prestigious National Accrediting Board of Higher Education (NAB) has recently rejected the University's application for membership citing lack of a comprehensive exam as a major reason. Accreditation by the NAB enhances a university's reputation to graduate schools, employers, and demonstrates to alumni that the school is worth supporting. A recent survey of influential alumni in corporations and the state legislature has revealed that contributions would improve significantly if the exams were instituted. With increased alumni support, continued increases in tuition might be avoided.

favorable, 35% unfavorable) when subjects are instructed to think about them, but weak messages should elicit a profile of predominately unfavorable thoughts (e.g., 65% unfavorable, 35% favorable) when subjects are instructed to scrutinize them. Since the thoughts elicited by particular arguments and messages might be expected to vary with different people, times, and contexts, pretesting is an essential aspect of message construction for any given experiment. For example, over the past 10 years we have found that students have generally become more knowledgeable about and accepting of the senior comprehensive exam idea. One consequence of this is that over the years we have had to develop arguments that are more specious in order to create empirically "weak" messages on this topic.

S4. A study conducted by the Educational Testing Service of Princeton, New Jersey, revealed that most of the Ivy League schools and several of the Big 10 universities have senior comprehensive exams to maintain their academic excellence. Professors at those schools who were interviewed recently said that senior comprehensive exams assured that only high quality and knowledgeable students would be associated with the university. This, of course, increases the prestige of current students, alumni of the school, and the university as a whole. The exams should be instituted to increase the academic reputation of the university. A national educator's publication recently predicted that within the next 10 years, the top universities would have the exam policy, and the weaker ones would not.

S5. An interesting and important feature of the comprehensive exam requirement is that it has led to a significant improvement in the quality of undergraduate teaching in the schools where it has been tried. Data from the Educational Testing Service confirm that teachers and courses at the schools with comprehensive exams were rated more positively by students after the exams than before. The improvement in teaching effectiveness appears to be due to departments placing more emphasis on high quality and stimulating teaching because departments look bad when their majors do poorly on the exam. For example, at the University of Florida, student ratings of courses increased significantly after comprehensive exams were instituted.

S6. One aspect of the comprehensive exam requirement that students at the schools where it has been tried seem to like is that all regular final examinations for seniors are typically eliminated. This elimination of final exams in all courses for seniors allows them to better integrate and think about the material in their major area just prior to graduation rather than "wasting" a lot of time cramming to pass tests in courses in which they are really not interested. Students presently have to take too many courses in subjects that are irrelevant to their career plans. The comprehensive exam places somewhat greater emphasis on the student's major and allows greater concentration on the material that the student feels is most relevant.

S7. Faculty members at universities with the comprehensive exams who were interviewed by researchers from the Carnegie Commission on Higher Education revealed that the comprehensive exams appeared to provide an incentive for students to study the material in their major area. A thorough study undertaken by the Department of Education at the University of Notre Dame showed that universities with comprehensive exams have resisted the national trend of declining scores on standardized achievement tests. Average scores on achievement tests for the universities with comprehensive exams have actually risen over the last five years.

S8. Data from the University of Virginia, where comprehensive exams were recently instituted, indicate that the average starting salary of graduates

increased over \$4000 over the two-year period in which the exams were begun. At comparable universities without comprehensive exams, salaries increased only \$850 over the same period. As Saul Siegel, a vice-president of IBM put it in *Business Week* recently, "We are much quicker to offer the large salaries and executive positions to these kids because by passing their area exam, they have proven to us that they have expertise in their area rather than being people who may or may not be dependable and reliable." Another benefit is that universities with the exams attract larger and more well-known corporations to campus to recruit students for their open positions. The end result is that students at schools with comprehensive exams have a 55% greater chance of landing a good job than students at schools without the exams.

S9. A study by the U.S. Department of Education revealed that universities with the comprehensive exam requirement average about 32% more financial aid available to students than comparable universities without the exams. Richard Collings, Director of Financial Aid at the University of Southern California (USC) has written that since the comprehensive exam was instituted at USC five years ago, more individuals and corporations have been willing to donate money for student scholarships.

Example Weak Arguments

W1. The National Scholarship Achievement Board recently revealed the results of a study they conducted on the effectiveness of comprehensive exams at Duke University. One major finding was that student anxiety had increased by 31%. At comparable schools without the exam, anxiety increased by only 8%. The Board reasoned that anxiety over the exams, or fear of failure, would motivate students to study more in their courses while they were taking them. It is likely that this increase in anxiety observed at Duke University would also be observed and be of benefit at other universities that adopt the exam policy.

W2. Graduate students have always had to take a comprehensive exam in their major area before receiving their degrees, and it is only fair that undergraduates should have to take them also. As the Dean of the Harvard Business School said, "If a comprehensive exam is considered necessary to demonstrate competence for a masters or doctoral degree, by what logic is it excluded as a requirement for the bachelors degree? What administrators don't realize is that this is discrimination just like discrimination against Blacks or Jews. There would be a lot of trouble if universities required only whites to take comprehensive exams but not Blacks. Yet universities all over the country are getting away with the same thing by requiring graduate students but not undergraduates to take the exams." Thus, the institution of comprehensive exams could be as useful for undergraduates as they have been for graduate students.

W3. A member of the Board of Curators has stated publicly that his brother had to take a comprehensive exam while in college and now he is manager of a large restaurant. He indicated that he realized the value of the exams since their father was a migrant worker who didn't even finish high school. He also indicated that the university has received several letters from parents in support of the exam. In fact, 4 of the 6 parents who wrote in thought that the exams were an excellent idea. Also, the prestigious National Accrediting Board of Higher Education seeks input from parents as well as students, faculty, and administrators when evaluating a university. Since most parents contribute financially to their child's education and also favor the exams, the university should institute them. This would show that the university is willing to listen to and follow the parents' wishes over those of students and faculty who may simply fear the work involved in comprehensive exams.

W4. A study conducted by the Educational Testing Service of Princeton, New Jersey revealed that many universities are considering adopting comprehensive exams. Thus, any university that adopted the exams could be at the forefront of a national trend. Some professors at schools with the exams who were interviewed felt that high school students would be impressed by a university that kept pace with current trends. In fact, whether or not a school had a comprehensive exam might be a determining factor in their choice of a university. Therefore, the enrollments of universities with the exams should increase as the information about the exams spreads among high school students.

W5. An interesting and important feature of the comprehensive exam requirement is that if the exams were instituted nationwide, students across the country could use the exam to compare their achievements with those of students at other schools. Data from the Educational Testing Service confirm that students are eager to compare their grades in a particular course with those of other students. Just imagine how exciting it would be for students in the Midwest to be able to compare their scores with those of students at the University of Florida, for example. This possibility for comparison would provide an incentive for students to study and achieve as high a score as possible so they would not be embarrassed when comparing scores with their friends.

W6. One feature of the comprehensive exam requirement that students at the schools where it has been tried seem to like is that passing the exams provides a very difficult challenge. For example, many students want jobs in business when they graduate and the corporate world is very tough. Yet, most students' lives are filled with few challenges whatsoever. Everything has been provided for them since the day they were born. It's not that students are not grateful, but knowing that they had to pass a difficult exam before they graduated would prepare them for the hard and cold realities of

life. Students would be nervous about passing the exam and fear that if they did not pass and graduate, four years of time would be wasted. However, that is what life is all about—taking risks and overcoming them. Having to pass a comprehensive exam is a challenge most students would welcome.

W7. Faculty members at universities with the comprehensive exams who were interviewed by researchers from the Carnegie Commission on Higher Education revealed that they liked the exams because it reduced the number of tests they felt they had to give in their classes knowing that students would still face one ultimate test of their knowledge in the comprehensive exam. A study at Norte Dame showed that this reduction in regular course tests saved enough paper to cover the cost of painting two classrooms.

W8. Data from the University of Virginia show that some students favor the senior comprehensive exam policy. For example, one faculty member asked his son to survey his fellow students at the school since it recently instituted the exams. Over 55% of his son's friends agreed that in principle, the exams would be beneficial. Of course, they didn't all agree but the fact that most did proves that undergraduates want the exams. As Saul Siegel, a student whose father is a vice-president of IBM wrote in the school newspaper: "The history of the exams can be traced to the ancient Greeks. If comprehensive exams were to be instituted, we could feel pleasure at following traditions begun by Plato and Aristotle. Even if there were no other benefits of the exams, it would be worth it just to follow tradition."

W9. A study by the U.S. Department of Education revealed that several national testing companies were developing comprehensive exams for use by universities in the U.S. The tests would be similar to the SAT and ACT tests which currently generate millions of dollars for the companies that make them. Richard Collings, a former Director of Financial Aid at the University of Southern California who now works for the Educational Testing Service, wrote recently in *Business Week*: "At ETS, we are not pushing comprehensive exams simply because of the huge amount of money involved. We are genuinely interested in marketing a good product. Just as our SAT and GRE tests are used to determine who is qualified for college and graduate work, so too should our comprehensive exams be used to determine who should graduate from college. We expect to have 32% of the market in 5 years."